

MARASANIGE SAMARTH MAHENDRA

+1 (857) 707-1671 | samarth.mahendragowda@gmail.com | Boston, MA, USA | LinkedIn | GitHub | Portfolio

EDUCATION

Northeastern University

January 2024 - December 2025

Master's, Computer Science

- Relevant coursework: Programming Design, Software Engineering, Database Systems, Computer Systems, Algorithms, NLP/ML, , Mobile development,

Dayananda Sagar College of Engineering

August 2018 - July 2022

Bachelor's, Computer Science

SKILLS

- Languages:** Python, Java, C/C++, SQL, NoSQL, JavaScript, TypeScript, HTML/CSS
- Frameworks & Libraries:** Django restframework, Flask, React.js, Node.js
- Databases:** PostgreSQL, Elasticsearch, MongoDB, Redis
- Cloud & DevOp:** AWS, Terraform, Docker, Kubernetes, Prometheus, Datadog, Celery
- Systems:** TCP/IP, Locking, Profiling, multi-threaded design, CPU microarchitecture, lock-free concurrency, cache optimization (L1/L2/L3), async I/O
- System Design & Concepts:** Microservices, REST APIs, System Design, scalable backend, Data Structures & Algorithms, distributed systems
- Testing:** Cypress, Jest, JUnit, PyTest
- Data structures and Algorithms:** Solved 400+ problems on LeetCode (Top 1.6 % of all LeetCode accounts (by global rank))

PROFESSIONAL EXPERIENCE

Jobstats.fyi

Remote

Founder & Developer

October 2025 - Present

- Built a real-time analytics platform visualizing 15K+ submissions from more than 3,700 candidates across 1K+ companies.
- Architected scalable Flask + MongoDB pipelines with Datadog RUM observability, reaching 1K+ sessions in 48 hours organically.

Draup

Bengaluru, KA, India

Associate Software development Engineer

August 2022 - November 2023

- Maintained and enhanced core platform modules across digital stack, outsourcing, customer, and university features, ensuring reliable operation and scalability.
- Designed a dynamic query framework enabling real-time data aggregation, improving chatbot performance by 60% and cutting new entity development time by 80%.
- Revamped filtering logic to support boolean modifiers operators and nested conditions, unlocking complex query composition for users.
- Developed 100+ modular REST APIs using Python, Django REST Framework, and PostgreSQL, with Redis caching and Celery workers for async task execution.
- Implemented subscription-based access control to manage app-level permissions and feature visibility.
- Led migration from PostgreSQL to Elasticsearch, enabling real-time analytics with 5x faster queries and higher scalability.
- Applied query optimization (partitioning, indexing, view creation) to achieve 400% faster execution and 50% lower operational cost.
- Built Datadog + AWS CloudWatch dashboards to monitor system health, reducing downtime from 4% to 1% and accelerating issue resolution by 75%.

Associate Software development Engineer Intern

April 2022 - July 2022

- Built Datadog dashboards, integrated AWS CloudWatch alarms to monitor platform health, reducing issue resolution time by 30%
- Implemented caching to improve efficiency of image requests, resulting in a 70% reduction in load times
- Developed self-running Jenkins jobs for database cleanup, cutting manual effort and improving efficiency by 25%

PROJECTS & OUTSIDE EXPERIENCE

ButterDB — High-Performance Buffered B-Tree Key-Value Store (C, Python)

Remote

Python, (C, Concurrency, WAL)

October 2025 - Present

- Built a concurrent single-node TCP key-value database with custom on-disk B-Tree engine and write buffering (Be-Tree).
- Implemented fixed-page file IO, per-node locking, Write-Ahead Logging (WAL), and crash recovery for durability and concurrency.
- Designed benchmarking and metrics collection to evaluate throughput, latency, and flush efficiency under mixed read/write workloads.
- Achieved ~8x higher insert throughput via batched buffer propagation and IO batching; code structured for systems-level performance experiments.

Open Jobs Analytics Platform – Backend Infra + Monitoring

Boston, MA, USA

Tech stack: Puppeteer, Python, Redis, celery, Mongodb, Grafana, Prometheus, GPT-4o

December 2024 - Present

- Designed a producer-consumer architecture using Celery, integrated with Prometheus and Grafana, achieving 99.9% uptime
- Scraped dynamic web pages with Playwright and Puppeteer, harvesting 1000+ data points daily.
- LLM-powered CSS selector extraction reduced new-site onboarding by 90%.
- Enhanced stealth capabilities with random headers, user agents, referrer headers, and OS configurations, reducing bot detection by up to 90%.
- [Link to project](#)

Real-Time AI Voice Assistant & Intelligent Agent Platform | OpenAI GPT-4o, Twilio, Discord, FastAPI

Boston, MA, USA

Tech stack: OpenAI GPT-4o, Twilio, Celery, FastAPI, Discord, Websockets, Render (deployment), Redis

April 2025 - May 2025

- Architected an agent system integrating OpenAI GPT-4 + Google Gemini with modular tools, dynamic function calling, and profile-aware responses via MongoDB and Discord.

- Built a scalable async backend with FastAPI + WebSockets, deployed on Render with a Celery worker handling long-running tool calls and real-time audio coordination.
- Deployed live demo via public phone number (833) 970-3274 using Twilio, showcasing job-query answering, system prompts with resume context, and cross-platform communication.
- [Link to project](#)

StackOverflow

February 2025 - April 2025

Full-Stack Q&A Platform (StackOverflow Clone) | TypeScript, JavaScript, React.js, Node.js, MongoDB, Cypress, Jest, CodeQL, DevOps

- Developed a full-stack Q&A system using React + Node.js/TypeScript with MVC backend, comprehensive Cypress/Jest testing, and CI/CD via GitHub Actions + CodeQL.
- [Link to project](#)

Myocardium Wall motion and wall thickness map, Patent Pending :- App-no : 202341086278 (India) (Co-inventor/author)

Bengaluru, KA, India

Research Assistant | Python, Dicom, MRI , numpy, Tkinter, image processing

November 2021 - September 2023

- Engineered a Python+DICOM cine-MRI processing system with custom wall-thickness algorithms, improving precision by 50% and accelerating runtime by 60× via NumPy + multiprocessing.